| | | | |
|---|---|---|---|
| **Fiscal Year:** | FY 2011 | **Task Last Updated:** | FY 02/03/2011 |
| **PI Name:** | Volz, Martin Ph.D. | | |
| **Project Title:** | Reduction of Defects in Germanium-Silicon (RDGS) | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | MATERIALS SCIENCE | | |
| **Program/Discipline--Element/Subdiscipline:** | MATERIALS SCIENCE--Materials science | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | martin.volz@nasa.gov | **Fax:** | FY |
| **PI Organization Type:** | NASA CENTER | **Phone:** | 256-544-5078 |
| **Organization Name:** | NASA Marshall Space Flight Center | | |
| **PI Address 1:** | EM31 | | |
| **PI Address 2:** | | | |
| **PI Web Page:** | | | |
| **City:** | Huntsville | **State:** | AL |
| **Zip Code:** | 35812 | **Congressional District:** | 5 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 96-HEDS-02 |
| **Start Date:** | 03/06/2008 | **End Date:** | 03/05/2015 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA MSFC |
| **Contact Monitor:** | Downey, James1 Patton | **Contact Phone:** | |
| **Contact Email:** | patton.downey@nasa.gov | | |
| **Flight Program:** | ISS | | |
| **Flight Assignment:** | NOTE: end date change per P. Downey/MSFC (Ed., 12/21/2011) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Mazuruk, Konstantin ( University of Alabama Huntsville ) | | |
| **Grant/Contract No.:** | | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| | NOTE: Original Principal Investigator was Dr. Frank Szofran, Feb 1998-Jan 2004; project continues with Dr. Martin Volz.<br>It is well established that crystals grown without contact with a container have far superior quality to otherwise similar crystals grown in direct contact with a container. In addition to float-zone processing, detached-Bridgman growth is often cited as a promising tool to improve crystal quality, without the limitations of float zoning. In detached growth, most of the melt has contact with the crucible wall, but the crystal is separated from the wall by a small gap, typically 10-100 microns. A liquid meniscus extends between the top edge of the growing crystal and the crucible wall. Detached growth has been found to occur quite often during microgravity experiments and considerable improvements of crystal quality have been reported for those cases. However, no thorough understanding of the process or quantitative assessment of the quality improvements existed prior to the investigation. This project will determine the means to reproducibly grow | | |

| | |
|---|---|
| **Task Description:** | Ge-Si alloys in the detached mode.<br><br>Ge-Si alloys are of technological interest because with lower defect density materials it would be possible to make thermal neutron and x-ray optics, faster devices than with silicon, higher efficiency solar cells, and higher thermopower devices. Although these alloys are model materials in the sense that these results should be applicable to many other materials, the major technological significance is that the same production equipment that is currently used for silicon device manufacture can also be used for germanium-silicon devices. This means that using germanium-silicon instead of some other material to achieve improved device performance will save an immense amount of money in development and purchase of production equipment.<br><br>Several tasks have been completed or are planned as part of the investigation:<br><br>* measure the relevant material parameters such as contact angle, growth angle, surface tension, and wetting behavior of the GeSi-melt on potential crucible materials;<br><br>* determine the mechanism of detached growth including the role of convection;<br><br>* quantitatively determine the differences in defects and impurities among normal Bridgman, detached Bridgman, and float zone (FZ) growth;<br><br>* investigate the influence of defined azimuthal or meridional flow due to rotating magnetic fields on the characteristics of detached growth;<br><br>* control time-dependent Marangoni convection in the case of FZ-growth by the use of a rotating magnetic field to examine the influence on the curvature of the solid-liquid interface and the heat and mass transport; and<br><br>* grow high quality GeSi single crystals with Si concentrations up to 10 at% and diameters up to 20 mm. Determine the necessary conditions for reproducibly achieving detachment of Ge-Si alloys (up to 10% Si) in the Bridgman growth method, and quantitatively compare the defect structures and impurity levels of detached Bridgman, normal Bridgman, and floating-zone grown materials. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The results of this research will increase our understanding of the mechanisms which lead to crystal detachment and enhance our ability to control detachment both in microgravity and on Earth. Control of detachment will enable the growth of relatively large diameter crystals with orders of magnitude fewer defects than can be achieved with standard growth techniques. The improvement in material quality may lead to better performance for a variety of electronic devices. |
| **Task Progress:** | A study was conducted to determine the conditions under which menisci can form in the detached crystal growth process. The shapes of menisci, under various conditions, were also calculated. In detached Bridgman growth, the crystal radius is less than the crucible radius and a meniscus bridges the gap between the crystal and crucible wall. Existence of detached growth depends upon the contact angle of the melt with the crucible wall, the growth angle of the solidifying crystal, the pressure differential across the meniscus, and the Bond number. The Young-Laplace capillary equation was used to calculate the crystal radii of detached states as a function of the pressure differential across the meniscus. Both terrestrial and microgravity conditions were considered. A variety of solution states were found, and multiple solutions with different crystal radii can exist for a given value of the pressure differential. The meniscus shapes for the detached states varied from simply convex or concave to shapes with numerous inflections. An approximate solution to the Young-Laplace equation for small Bond numbers was derived. Some of the results of this study have been submitted for publication.<br>Another study undertaken in the reporting period concerns the static stability of menisci in detached Bridgman growth. The Young-Laplace equation can be used to calculate detached states but the only states that are physically realizable are those which are statically stable. Stable states are those for which the potential energy is at a minimum. The meniscus stability analysis consists of determining the sign of the second-order correction to the energy for any admissible meniscus perturbations. Axisymmetric as well as non-axisymmetric perturbations have been investigated. Both rough and microscopically smooth crucible surfaces, corresponding to pinned and unpinned states, have also been considered. The analysis of static stability is continuing and results are expected to be submitted for publication in the next reporting period. |
| **Bibliography Type:** | Description: (Last Updated: 07/09/2021) |
| **Abstracts for Journals and Proceedings** | Szofran FR, Chiaramonte FP, Kelton KF, Matson DM, Poirier DR, Trivedi RK, Su CH, Volz MP, Voorhees PW. "U.S. Materials Science on the International Space Station." Presented at the 48th AIAA Aerospace Sciences Meeting, Orlando, Florida, January 4-7, 2010.<br>48th AIAA Aerospace Sciences Meeting Abstracts, January 2010. , Jan-2010 |
| **Abstracts for Journals and Proceedings** | Volz MP, Mazuruk K. "Meniscus Shapes in Detached Bridgman Growth." Presented at the 16th International Conference on Crystal Growth, Beijing, China, August 8-13, 2010.<br>16th International Conference on Crystal Growth, August 2010. , Aug-2010 |
| **Abstracts for Journals and Proceedings** | Mazuruk K, Volz MP. "Linear Stability of Binary Alloy Solidification for Unsteady Growth Rates." 16th International Conference on Crystal Growth, Beijing, China, August 8-13, 2010.<br>16th International Conference on Crystal Growth, August 2010. , Aug-2010 |